

Nemeth Braille

The **Nemeth Braille Code for Mathematics and Science Notation** is a Braille code for encoding mathematical and scientific notation linearly using standard six-dot Braille cells for tactile reading by the visually impaired. The code was developed by Abraham Nemeth. The Nemeth Code was first written up in 1952. It was revised in 1956, 1965, and 1972.^[1] It is an example of a compact human-readable markup language.

Nemeth Braille is just one code used to write mathematics in braille. There are many systems in use around the world.^[2]

Nemeth Braille	
Script type	<u>Alphabet Braille code</u> for encoding mathematical and scientific notation linearly
Creator	<u>Abraham Nemeth</u>
Time period	1952 to the present
Languages	Several

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Principles of the Nemeth Code

The Nemeth Code Book (1972) opens with the following words:

This Braille Code for Mathematics and Science Notation has been prepared to provide a system of symbols which will allow technical literature to be presented and read in braille. The Code is intended to convey as accurate an impression as is possible to the braille reader of the corresponding printed text, and this is one of its principal features. When the braille reader has a clear conception of the corresponding printed text, the area of communication between himself and his teacher, his colleagues, his associates, and the world at large is greatly broadened. *A test of the accuracy with which the Code conveys information from the print to the braille text is to effect a transcription in the reverse direction. The amount of agreement between the original printed text and one transcribed from the braille is a measure of the Code's accuracy.*

—[3]

One consequence is that the braille transcriber does not need to know the underlying mathematics. The braille transcriber needs to identify the inkprint symbols and know how to render them in Nemeth Code braille. For example, if the same math symbol might have two different meanings, this would not matter; both instances would be brailled the same. This is in contrast to the International Braille Music Code, where the braille depends on the meaning of the inkprint music. Thus a knowledge of music is required to produce braille music.

Table of Nemeth braille codes

General signs

Symbol	*	,	...
<u>Braille</u>	⠠ ⠨ ⠠ ⠨ ⠠ ⠨ ⠠ ⠠ ⠠ ⠠ ⠠ ⠠ ⠠ ⠠ ⠠ ⠠ ⠠ ⠠	⠠ ⠨ ⠠ ⠨ ⠠ ⠨ ⠠ ⠠ ⠠ ⠠ ⠠ ⠠ ⠠ ⠠ ⠠ ⠠ ⠠ ⠠	⠠ ⠨ ⠠ ⠨ ⠠ ⠨ ⠠ ⠠ ⠠ ⠠ ⠠ ⠠ ⠠ ⠠ ⠠ ⠠ ⠠ ⠠

[4]

Number signs

Symbol	Number prefix	Decimal point	Thousands separator	0	1	2	3	4	5	6	7	8	9	
Braille														

[3]

Operators

Symbol	+	-	×	.	÷	. (decimal)	√	i√ (radical index)
Braille								
Symbol	inner √ (1st)	inner √ (2nd)	end 1st √	end 2nd √	long division	/	!	
Braille								

[4]

Parentheses and brackets

Symbol	(	)	[	]	{	}	
Braille							

[4]

Fractions

Symbol	simple			complex			hypercomplex			fraction in mixed number			
	open	line	close	open	line	close	open	line	close	open	line	close	
Braille													

[4]

Other modifiers

Symbol	∴	∴	:	∴	'	”
<u>Braille</u>						

[4]

Symbol	∞	∝	°	%	√
Braille					

[4]

Braille indicators

Symbol	Punctuation	Superscript	Subscript	Baseline	Omission	Cancel (close)	Cancel (open)	Run Over	Directly Over	Directly Under	Beginning	Termination
Braille												

[4]

Comparison signs

Symbol	>	<	≥	≤	≧	≦	
Braille	• • •	• • •	• • •	• • •	• • •	• • •	• • •

[4]

Symbol	$\geq$	$\leq$	$\geq$	$\leq$
Braille	• ◻ • ◻ • ◻ ◻ ◻ ◻ • ◻ ◻ • ◻ ◻ ◻ ◻ • ◻ • ◻ ◻ ◻ ◻	• ◻ ◻ ◻ ◻ • ◻ ◻ ◻ • ◻ ◻ • ◻ ◻ ◻ ◻ • ◻ • ◻ ◻ ◻ ◻	◻ • • ◻ ◻ ◻ ◻ ◻ ◻ ◻ ◻ ◻ ◻ ◻ • ◻ ◻ • • ◻ ◻ ◻ • ◻	◻ ◻ • • ◻ ◻ ◻ ◻ ◻ ◻ ◻ ◻ ◻ ◻ • ◻ ◻ • • ◻ ◻ ◻ ◻ •

[4]

Symbol	><	<>	>=<	<=>
Braille	• • • • • • • • •	• • • • • • • • •	• • • • • • • • • •	• • • • • • • • • •

[4]

Symbol	=	≠	≻	≺	~	≈	≍		
Braille	• • • •	• • • •	• •	• •	•	• •	• •	• •	

[4]

Geometry

Symbol	Parallel	Perpendicular	Angle	Right angle
Braille	● ● ● ○ ○ ○ ● ○ ○ ● ○ ○ ○ ● ○ ● ○ ○	● ● ● ● ○ ○ ○ ○ ○ ● ○ ○ ○ ● ● ○ ○ ○	● ● ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○	● ● ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○

[4]

Symbol	Arc down	Arc up	Trapezoid	Square	Rectangle	Parallelogram
Braille	● ● ● ○ ○ ○ ● ○ ○ ○ ○ ○ ○ ● ○ ○ ○ ○	● ● ● ○ ○ ○ ● ○ ○ ○ ○ ○ ○ ● ● ○ ○ ○	● ● ● ● ● ● ● ○ ○ ● ● ● ○ ○ ○ ○ ○ ○	● ● ● ○ ○ ○ ● ○ ○ ● ● ● ○ ○ ○ ○ ○ ○	● ● ● ○ ○ ○ ● ○ ○ ● ● ● ○ ○ ○ ○ ○ ○	● ● ● ● ● ● ● ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○

[4]

Symbol	Rhombus	Isosceles trapezoid	Triangle	Right triangle	Circle	Ellipse
Braille	●●●○ ●○●● ○●○○	●●●○ ●○ ●● ○● ●●	●●●○ ●○●● ○●●○	●● ○● ●● ○●●● ●○ ●● ○● ●● ●● ○● ●● ○● ●● ○● ○●	●●●● ●○ ○● ●● ●● ○● ○● ○● ○●	●●●○ ●○ ○● ●● ○● ○● ○●

[4]

Arrows

Symbol	→	←	↔
Braille	• • ○ ○ ○ ○ • ○ • ○ • • • • ○ • ○ • ○ ○ ○ ○ • ○	• • ○ • ○ ○ ○ ○ • ○ • ○ • • • • ○ • ○ • ○ ○ ○ ○	• • ○ • ○ ○ ○ ○ • ○ • • • • ○ • ○ • ○ • ○ ○ ○ ○

[4]

Symbol	→	⇌	⇒
Braille	• • • • • • • • • •	• • • • • • • • • • • • • • • • • •	• • • • • • • • • • • • • • • • • • • • • • • • • •

[4]

Function	Sine	Cosine	Tangent
<u>Braille</u>			

Function	Secant	Cosecant	Cotangent	Logarithm
<u>Braille</u>				

Symbols	$\in$	$\ni$	$\cup$	$\cap$	$\emptyset$	$\{$	$\}$
Braille							

Symbol	$\subset$	$\supset$	$\subseteq$	$\supseteq$
<u>Braille</u>	⠠⠨⠠⠠⠠⠠⠠ ⠠⠨⠠⠠⠠⠠⠠ ⠠⠨⠠⠠⠠⠠⠠	⠠⠨⠠⠠⠠⠠⠠ ⠠⠨⠠⠠⠠⠠⠠ ⠠⠨⠠⠠⠠⠠⠠	⠠⠨⠠⠠⠠⠠⠠⠠⠠ ⠠⠨⠠⠠⠠⠠⠠⠠⠠ ⠠⠨⠠⠠⠠⠠⠠⠠⠠	⠠⠨⠠⠠⠠⠠⠠⠠⠠ ⠠⠨⠠⠠⠠⠠⠠⠠⠠ ⠠⠨⠠⠠⠠⠠⠠⠠⠠

Symbol	$\oplus$	$\otimes$	$\square$
<u>Braille</u>			

Symbol	Intersecting lines	@	Ditto	Tally mark	+-	÷
<u>Braille</u>						

Symbol		≈	≡	—	(modulus)	. dot
<u>Braille</u>	⠇ ⠏	⠁ ⠠ ⠠ ⠠ ⠠ ⠠	⠁ ⠠ ⠠ ⠠ ⠠ ⠠	⠠ ⠏	⠠ ⠠ ⠠ ⠠ ⠠ ⠠	⠠ ⠠ ⠠

Symbol	Pentagon	Hexagon	Heptagon	Octagon	Nonagon	n -agon
<u>Braille</u>						+ (Nemeth Braille of n)

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Triangles

Symbol	Acute	Right	Isosceles	Obtuse	Scalene
Braille					

[4]

Other signs

Symbol	Acute angle	Right angle	Obtuse angle	Hexagon star	¶	§
Braille						

[4]

Symbol	ref. indicator	<i>		letter sign	composite function	det matrix	∫	∂
Braille								

[4]

Greek letters and Latin letters

Greek and Latin letters are based on the assignments of International Greek Braille.

Symbol	small Latin letters	capital Latin letters	small Greek letters	capital Greek letters
Braille	(Braille of the letter)			

[4]

See also

- Gardner-Salinas braille
- WIMATS, application software to transcribe mathematical and scientific text input into braille script.

References

1. "The History of Nemeth Code" (<http://nfb.org/images/nfb/publications/fr/fr28/fr280110.htm>). Retrieved 2012-04-26.

2. "Braille Mathematical Notations" (http://chezdom.net/blog/?page_id=51). 2008-07-22. Retrieved 2012-04-26.

3. "The Nemeth Braille Code for Mathematics and Science Notation 1972 Revision" (<http://www.brailleauthority.org/mathscience/nemeth1972.pdf>) (PDF). Retrieved 2012-04-26.

4. "Nemeth Braille (Mathematics Braille)" (https://web.archive.org/web/20120421113105/http://acharya.iitm.ac.in/disabilities/nemeth_brl.php). Archived from the original (http://acharya.iitm.ac.in/disabilities/nemeth_brl.php) on 2012-04-21. Retrieved 2012-04-25.

External links

- An Introduction to the Nemeth Braille Code for Mathematics (<http://www.dotlessbraille.org/NemethIntro.htm>)
- PDF of the 1972 Nemeth Braille Code for Mathematics (<http://www.brailleauthority.org/mathscience/nemeth1972.pdf>)
- latex2nemeth (<http://myria.math.aegean.gr/braille/index-en.html>), application software that transcribes LaTeX to Nemeth Braille.
- Braille files for the Nemeth Braille Code for Mathematics; and other resources about braille transcribing (<http://www.duxburysystems.org/documentation/braille/index.htm>)
- The History of Nemeth Code (<https://nfb.org/images/nfb/publications/fr/fr28/fr280110.htm>)
- Braille Mathematical Notations: Different Braille Mathematical Codes from Around the World (<https://chezdom.net/mathematicalbraillecodes/>)

Источник — https://en.wikipedia.org/w/index.php?title=Nemeth_Braille&oldid=1222781245

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